FABRIC: Federated Adaptive BiomedCLIP with Instance-level Clinical Prompts

Dynamic Prompt Enabled BioMedCLIP in a Federated Learning Setup

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Abstract

Federated learning (FL) has demonstrated strong potential for privacy-preserving medical image analysis; however, adapting large-scale vision-language models (VLMs) to heterogeneous clinical data remains challenging due to communication overhead, domain shift, and limited prompt flexibility. We present **FABRIC** (**F**ederated **A**daptive **B**iomedCLIP with **I**nstance-level **C**linical prompts), an enhanced federated learning framework that extends FACMIC [19] along four dimensions: replacing CLIP with BiomedCLIP for domain-aware feature extraction, substituting static text prompts with image-level clinical descriptions generated by Llama 3.2 via Ollama, adopting an asynchronous FL protocol where only two of three clients participate per round, and incorporating class-aware offline data augmentation. Experiments on brain tumor MRI classification show that our method achieves mean ACC 87.68% and BACC 88.05% under leave-one-client-out evaluation, outperforming FACMIC by +4.06% ACC and +5.52% BACC on the global test set.

Keywords: FABRIC, BioMedCLIP, Dynamic Prompts, Federated Learning, Data Augmentation, Asynchronous FL

1. Introduction

The success of deep learning in medical image analysis depends on access to large, well-annotated datasets. In clinical settings, this is severely limited by privacy regulations such as GDPR [18] and HIPAA [17], which prohibit sharing raw patient data across institutions. Federated learning (FL) [3] addresses this by enabling collaborative model training where only model parameters, not data, are communicated. Despite its promise, FL in the medical domain faces persistent challenges of statistical heterogeneity and communication cost [16].

FACMIC [19] made an important contribution by adapting CLIP-based vision-language models to this federated setting with lightweight attention modules and domain adaptation. However, it relies on a general-purpose CLIP backbone that was not designed for biomedical data, uses fixed text prompts that carry no clinical context, requires all clients to participate every round, and does not address class imbalance through augmentation. These limitations leave significant room for improvement, particularly in terms of representation quality and training robustness. We therefore introduce **FABRIC** to address these gaps.

2. Problem Statement

We consider a federated learning scenario with N clients $\mathcal{C} = \{C_1, \dots, C_N\}$, where each client C_i holds a private dataset $\mathcal{D}_i = \{(\mathbf{x}_{i,j}, y_{i,j})\}_{j=1}^{n_i}$, with $\mathbf{x}_{i,j} \in \mathcal{X}$ a medical image and $y_{i,j} \in \mathcal{Y} = \{1, \dots, K\}$ its class label. Following standard heterogeneous FL assumptions [16], clients share the same input and output spaces but follow different data distributions:

$$P(\mathcal{D}_i) \neq P(\mathcal{D}_{i'}), \quad \forall\, i \neq i'. \tag{1}$$

Raw data never leave any client; only model parameters are communicated to the central server. Each dataset $\mathcal{D}_i$ is further partitioned into non-overlapping training, validation, and test subsets. The global objective is to learn a model $f_{\boldsymbol{\theta}} : \mathcal{X} \to \mathcal{Y}$ minimising the expected classification loss across all clients:

$$\min_f \; \frac{1}{N} \sum_{i=1}^{N} \frac{1}{n_i^{\text{test}}} \sum_{j=1}^{n_i^{\text{test}}} \ell\big(f_{\boldsymbol{\theta}}(\mathbf{x}_{i,j}^{\text{test}}), y_{i,j}^{\text{test}}\big)\,, \tag{2}$$

where $\ell(\cdot,\cdot)$ is the task loss function. In a vision-language framework, classification is performed by comparing image embeddings against text embeddings derived from class-specific prompts, without accessing raw data from other clients.

The existing approach FACMIC [19] tackles distributional shift via a feature attention module and LMMD domain adaptation, but uses CLIP [15] as its backbone — a general-purpose model pre-trained on web data with no biomedical content — and relies on a fixed text prompt `"a picture of a {class}"` that provides no clinical or modality-specific context. We address both of these shortcomings and evaluate under both the standard `client_3` protocol and a leave-one-client-out (LOCO) protocol, where each client is designated as the test set in turn.

3. Challenges in existing work

Challenges Prior to FACMIC

Before FACMIC, the dominant approach to applying large VLMs such as CLIP in FL was to fine-tune and aggregate the full model weights across clients each round [10]. With CLIP's ViT-B/32 backbone exceeding 10^8 parameters, this was prohibitively expensive in bandwidth-limited clinical networks, making per-round communication a critical bottleneck. Moreover, standard FL algorithms such as FedAVG, FedProx [9], and MOON [8] were not designed for vision-language models and could not exploit the rich semantic representations that CLIP provides. Distributional shift across clients — arising from differences in imaging equipment, acquisition protocols, and patient demographics — further degraded global model quality, and no existing method addressed both the communication and domain shift problems simultaneously within a VLM-based FL framework.

Why FACMIC Was Introduced

FACMIC [19] was proposed to address these limitations by keeping both the CLIP image and text encoders frozen and training only a lightweight five-layer feature attention module per client. This reduced the number of communicated parameters from over 10^8 (full CLIP) to a small fraction, making VLM-based FL practically feasible for the first time. FACMIC further introduced a Local Maximum Mean Discrepancy (LMMD) domain adaptation loss to align class-conditional feature distributions between each client's local data and a shared set of unlabelled target images, addressing the distributional shift problem. The use of a fixed text prompt `"a picture of a {class}"` combined with CLIP's zero-shot classification ability enabled classification without any label sharing between clients. Despite these contributions, several important limitations remain, which FABRIC directly addresses.

4. Methodology

FACMIC: Author's Proposed Methodology

FACMIC [19] addresses communication cost and distributional shift in VLM-based FL through three tightly coupled components: a frozen CLIP backbone that provides rich pre-trained representations, a client-specific feature attention module that selects task-relevant features, and an LMMD-based domain adaptation strategy that aligns feature distributions across clients.

4.1 FACMIC Architecture

FACMIC uses a pre-trained CLIP model with a frozen image encoder $e_I(\cdot)$ and text encoder $e_T(\cdot)$. For a training image $\mathbf{x}_j \in \mathcal{D}_i^{\text{train}}$, the image encoder extracts a D-dimensional feature vector $\mathbf{I}_j = e_I(\mathbf{x}_j) \in \mathbb{R}^D$. Text features $\mathbf{T}_c = e_T(p_c) \in \mathbb{R}^D$ are obtained by encoding the fixed class prompt `"a picture of a {class}"` through the text encoder. A lightweight client-specific attention module $a_i(\cdot)$ takes the image features as input and produces a soft attention mask $a_i(\mathbf{I}_j) \in [0,1]^D$, applied element-wise to yield masked image features:

$$\tilde{\mathbf{I}}_j = a_i(\mathbf{I}_j) \odot \mathbf{I}_j, \tag{3}$$

where $\odot$ denotes the Hadamard product. This masking operation allows each client to selectively emphasise the feature dimensions most relevant to its local data distribution, effectively personalising the pre-trained representation without modifying the frozen backbone. The attention module has five layers: a linear projection , batch normalisation, LeakyReLU activation, a second linear projection , and a softmax output that normalises the mask to $[0,1]^D$. Since only the attention module parameters are communicated to the server at each round, the communication cost is reduced to a tiny fraction of the full CLIP model size.

Classification is performed via cosine similarity between the masked image features and the class-level text embeddings:

$$P(Y{=}c \mid \mathbf{x}_j) = \frac{\exp(s_{j,c}/\tau)}{\sum_{c'=1}^{K} \exp(s_{j,c'}/\tau)}, \quad s_{j,c} = \frac{\langle \tilde{\mathbf{I}}_j, \mathbf{T}_c \rangle}{\|\tilde{\mathbf{I}}_j\| \cdot \|\mathbf{T}_c\|}, \tag{4}$$

where τ is a learnable temperature parameter that controls the sharpness of the softmax distribution, and $s_{j,c}$ is the cosine similarity between the masked image embedding and the text embedding of class c. The predicted class is $\hat{y} = \arg\max_c P(Y{=}c \mid \mathbf{x}_j)$.

4.2 Contrastive Loss

The attention module parameters are optimised using a symmetric contrastive loss [15], which encourages the model to bring the image embeddings of a training sample close to its corresponding text embedding while pushing them away from the embeddings of all other classes in the batch. This objective ensures that the attention module learns to produce feature representations that are well-aligned with the semantic content described by the text prompts.

For a mini-batch of B training examples, let $\mathbf{S} \in \mathbb{R}^{B\times B}$ be the matrix of pairwise cosine similarities between all masked image features and text embeddings. The image-to-text probability matrix is $\mathbf{P} = \text{softmax}(\mathbf{S}/\tau) \in [0,1]^{B\times B}$ and the text-to-image probability matrix is $\mathbf{Q} = \text{softmax}(\mathbf{S}^\top/\tau) \in [0,1]^{B\times B}$. The symmetric contrastive loss is:

$$\mathcal{L}_{\text{contr}} = -\frac{1}{B}\sum_{j=1}^{B} \frac{1}{2}\big(\log p_{j,j} + \log q_{j,j}\big), \tag{5}$$

where $p_{j,j}$ and $q_{j,j}$ are the diagonal entries of $\mathbf{P}$ and $\mathbf{Q}$, corresponding to the probability that the j-th image is correctly matched to its own text embedding (and vice versa). Minimising this loss simultaneously maximises image-to-text and text-to-image alignment for all matched pairs in the batch, producing a richer and more balanced contrastive signal than a unidirectional objective.

4.3 LMMD Domain Adaptation Loss

In heterogeneous FL settings, the data distributions across clients can differ significantly. Training on local data alone may cause the global model to converge to a solution that performs well on some clients but poorly on others. To address this, FACMIC employs Local Maximum Mean Discrepancy (LMMD) [21], a class-conditional domain adaptation method that explicitly aligns the feature distributions of the source domain $\mathcal{D}^s$ (the client's local data) and the target domain $\mathcal{D}^t$ (a shared set of publicly available unlabelled images) within a reproducing kernel Hilbert space $\mathcal{H}$.

Unlike standard MMD, which aligns marginal distributions, LMMD performs *class-conditional* alignment, ensuring that features from the same class are aligned across domains rather than merely aligning the overall feature distributions. Applying the kernel trick with a Gaussian kernel $k(\cdot,\cdot)$, the domain adaptation loss is:

$$\mathcal{L}_{\mathrm{DA}} = \frac{1}{K}\sum_{c=1}^{K}\Big[\sum_{i,j}^{n_s}\omega_{i,c}^{s}\omega_{j,c}^{s}k(\mathbf{I}_i^s,\mathbf{I}_j^s) + \sum_{i,j}^{n_t}\omega_{i,c}^{t}\omega_{j,c}^{t}k(\mathbf{I}_i^t,\mathbf{I}_j^t) - 2\sum_{i}^{n_s}\sum_{j}^{n_t}\omega_{i,c}^{s}\omega_{j,c}^{t}k(\mathbf{I}_i^s,\mathbf{I}_j^t)\Big], \quad (6)$$

where n_s and n_t are the number of source and target samples, $\omega_{i,c}^{s|t} = \mathbf{1}(y_i{=}c)/\sum_j \mathbf{1}(y_j{=}c)$ are class-membership weights that measure how strongly sample i belongs to class c, and the kernel bandwidth is set to the median pairwise squared distance [21]. Since the target domain $\mathcal{D}^t$ is unlabelled, pseudo-labels are assigned by taking the highest-probability class from Eq. (4). The Gaussian kernel evaluates the similarity between image features in the source and target domains, with the three terms measuring intra-source similarity, intra-target similarity, and cross-domain similarity, respectively. Minimising $\mathcal{L}_{\mathrm{DA}}$ reduces the discrepancy between the source and target feature distributions class by class.

The total client training objective combines the contrastive and domain adaptation losses:

$$\mathcal{L} = \mathcal{L}_{\mathrm{contr}} + \lambda\mathcal{L}_{\mathrm{DA}}, \quad \lambda = 1, \quad (7)$$

where λ controls the trade-off between cross-modal alignment and domain shift reduction.

4.4 FACMIC Global Aggregation

At each global round, *all* N clients upload their locally trained attention module parameters $\boldsymbol{\theta}_i^t$ to the central server. The server combines them into a single global attention module via sample-weighted FedAVG [10]:

$$\boldsymbol{\theta}_{\mathrm{global}}^{t+1} = \sum_{i=1}^{N}\omega_i\boldsymbol{\theta}_i^t, \quad \omega_i = \frac{n_i^{\mathrm{train}}}{\sum_{i'} n_{i'}^{\mathrm{train}}}, \quad (8)$$

where ω_i is the fraction of total training samples held by client C_i, giving larger clients proportionally more influence on the global model. The aggregated parameters are then broadcast back to all clients for the next round. Since only the compact attention module (not the full CLIP encoders) is communicated, the per-round bandwidth cost is minimal regardless of backbone size.

4.5 Our proposed novelty : Improvements over FACMIC limitations

Improvement 1 — BiomedCLIP backbone. CLIP [15] is pre-trained on ∼400M web image-text pairs that are dominated by natural photographs and consumer content, with negligible representation of clinical imaging data. As a result, its image encoder lacks sensitivity to pathological visual features such as lesion morphology, tissue texture, and modality-specific artefacts that are critical for medical diagnosis. Furthermore, its text encoder was never exposed to clinical or anatomical terminology, producing poorly calibrated embeddings for disease-specific descriptions such as *glioma*, *meningioma*, or *basal cell carcinoma*. This domain gap is quantified by the finding that zero-shot CLIP achieves only 52.7% recall on the ISIC2019 skin cancer benchmark [14].

We replace CLIP with **BiomedCLIP** [20], a dual-encoder vision-language model comprising a Vision Transformer (ViT) [4] image encoder and a PubMedBERT [7] text encoder. PubMedBERT is pre-trained from scratch exclusively on full-text PubMed articles and abstracts, giving it a strong prior over biomedical vocabulary, anatomical structures, and clinical findings. BiomedCLIP is pre-trained on **PMC-15M** — 15 million biomedical figure-caption pairs from PubMed Central — using the same symmetric contrastive objective as Eq. (5). This training data directly reflects the kinds of medical images and clinical descriptions encountered in our task, yielding image features that are sensitive to pathological morphology and text embeddings that are well-calibrated for medical class names. Both BiomedCLIP encoders remain frozen throughout federated training, so replacing CLIP with BiomedCLIP introduces no additional communication cost: only the lightweight attention module continues to be communicated and aggregated.

Improvement 2 — Dynamic prompt generation. FACMIC applies the fixed template `"a picture of a {class}"` uniformly to all images, all clients, and all training rounds. This static formulation has two fundamental weaknesses. First, it conveys no modality-specific information: the prompt is identical regardless of whether the image is an MRI scan, a dermoscopic photograph, or a histological slide, giving the text encoder no signal about the imaging context. Second, for closely related disease categories — such as glioma versus meningioma, or melanoma versus pigmented benign keratosis — the fixed template produces nearly identical text embeddings, since the only difference is the class name appended to the same generic prefix. This severely weakens the contrastive signal, making it difficult for the attention module to learn features that discriminate between similar-looking pathologies.

We address this by introducing an *offline, image-level dynamic prompt generation* strategy using **Llama 3.2** [12], an open-weight large language model with strong biomedical knowledge, deployed locally via **Ollama** [13]. Ollama is a lightweight, open-source inference runtime that manages model quantisation and inference through a simple local `ollama.Client` API, routing all queries to an on-device server without any network calls to external endpoints. This design is critical for federated healthcare settings: no patient data, image content, or class information is transmitted to third parties, and no internet connectivity is required during training. For each image $\mathbf{x}_j$ of class c, Llama 3.2 is queried with a structured, role-conditioned instruction:

> *"You are a radiologist. Look at this brain MRI image which shows {class}. Generate a single concise medical description (one sentence, max 15 words) describing the key radiological findings. Focus on:*

location, borders, enhancement pattern, surrounding tissue effects."

The radiologist persona grounds the model's output in clinical terminology; the 15-word limit ensures compatibility with BiomedCLIP's token budget and avoids truncation. A fallback is applied when inference fails due to occasional model errors:

$$p_j = \begin{cases} \text{Llama 3.2 response,} & \text{if inference succeeds,} \\ \texttt{"brain MRI showing } c\texttt{",} & \text{otherwise.} \end{cases} \tag{9}$$

The fallback is semantically equivalent to the FACMIC template, ensuring training is never interrupted by a failed LLM call. Prompts are generated for *all* images — original and augmented — and serialised to a JSON file indexed by image path before training begins, adding zero LLM latency to the training loop. Importantly, each augmented copy $\mathbf{x}_j^{(l)}$ receives its own independently generated prompt $p_j^{(l)}$ rather than inheriting the prompt of its source image $\mathbf{x}_j$, since geometric transforms such as random cropping may shift the lesion to a different spatial location, subtly altering the radiologically relevant context. The image-specific prompt is encoded by the frozen BiomedCLIP text encoder:

$$\mathbf{T}_j = e_T^{\text{Biomed}}(p_j) \in \mathbb{R}^D, \tag{10}$$

which replaces the single class-level text embedding used in FACMIC in Eq. (4), producing an instance-aware contrastive signal grounded in clinically specific radiological descriptions.

Our Additional Experimental Contributions

Contribution A — Asynchronous federated learning. We adopt **asynchronous FL**, where at each global round t, a random subset $\mathcal{S}^t \subset \mathcal{C}$ of $|\mathcal{S}^t| = 2$ clients is selected to participate using a dedicated random number generator (seeded at 42, independent of data shuffling). The skipped client does not train and does not contribute parameters that round. The aggregation becomes:

$$\boldsymbol{\theta}_{\text{global}}^{t+1} = \sum_{i \in \mathcal{S}^t} \omega_i \boldsymbol{\theta}_i^t, \quad \omega_i = \frac{n_i^{\text{train}}}{\sum_{i' \in \mathcal{S}^t} n_{i'}^{\text{train}}}, \tag{11}$$

reducing per-round communication by one third compared to synchronous FL while improving robustness to client unavailability.

Contribution B — Data augmentation. We apply an **offline, class-aware augmentation pipeline** to training clients (`client_0`–`client_2`) before federated training. `client_3` is never augmented. Each image receives m_c distinct augmented copies:

$$m_c = \begin{cases} 3, & c = \texttt{no\_tumor} \ (\times 4 \text{ total}), \\ 2, & \text{otherwise} \ (\times 3 \text{ total}). \end{cases} \tag{12}$$

We define a pool of ten MRI-appropriate transforms $\mathcal{T} = \{T_1, \ldots, T_{10}\}$ (Table 1), with hue and saturation jitter excluded since MRI carries no colour information. The l-th augmented copy is:

$$\mathbf{x}_j^{(l)} = T_{k_l}(\mathbf{x}_j), \quad k_i \neq k_{i'} \ \forall i \neq i', \quad l = 1, \ldots, m_c, \tag{13}$$

with distinct transforms drawn without replacement, guaranteeing that no two copies of the same image share the same perturbation. All augmentations are saved offline (JPEG, quality 95) . Images are resized to 224×224 and normalised via z-score [5]. Each augmented copy receives its own Llama 3.2 prompt.

Table 1. Data augmentation transform pool ($\mathcal{T}$).

ID	Transform	Parameters
T_1	Horizontal flip	$p = 1.0$
T_2	Vertical flip	$p = 1.0$
T_3	Random rotation	$\pm 20°$, zero-fill
T_4	Random affine	$\pm 15°$, translate $(0.10, 0.10)$, scale $(0.85, 1.15)$, shear $10°$
T_5	Random resized crop	512×512, scale $(0.75, 1.0)$, ratio $(0.9, 1.1)$
T_6	Gaussian blur	Kernel 5×5, $\sigma \in (0.5, 2.0)$
T_7	Brightness/contrast	± 0.15 each
T_8	Flip + rotation	$T_1 \circ T_3$ $(\pm 15°)$
T_9	Affine + blur	Affine $(\pm 10°$, trans. $0.05) \circ$ blur $(3 \times 3$, $\sigma \in (0.3, 1.5))$
T_{10}	Crop + contrast	T_5 (scale 0.80–1.0) $\circ$ contrast ± 0.20

Contribution C — Comprehensive evaluation. FACMIC evaluates only on the single held-out `client_3` partition, which may conceal per-client performance variability under non-iid conditions. We additionally adopt a **leave-one-client-out** (LOCO) protocol, designating each of the four clients as the test client in turn while training on the remaining three, providing a complete picture of model generalisation across all client distributions.

4.6 Framework Overview

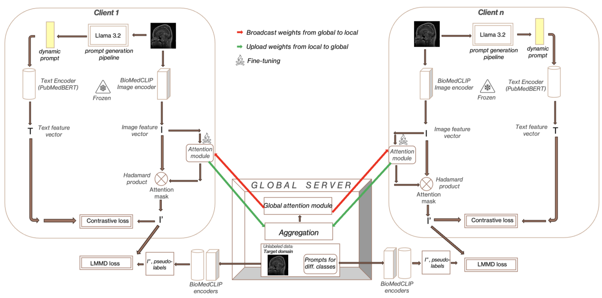

Figure 1. FABRIC framework. Each client trains its local attention module using contrastive and LMMD losses with BiomedCLIP features and Llama 3.2 dynamic prompts. Server aggregates via sample-weighted FedAVG and broadcasts back. Red: broadcast; Green: upload; Snowflake: frozen; Flame: fine-tuned (attention only).

Figure 1 illustrates the complete FABRIC framework. Each client's MRI image is encoded by the frozen BiomedCLIP image encoder to produce image features $\mathbf{I}_j$. Simultaneously, Llama 3.2 via Ollama generates an image-specific dynamic prompt, which is encoded by the frozen PubMedBERT text encoder to produce $\mathbf{T}_j$. The trainable attention module produces masked features $\tilde{\mathbf{I}}_j$ (Eq. (3)), used to compute $\mathcal{L}_{\text{contr}}$ (Eq. (5)) and $\mathcal{L}_{\text{DA}}$ (Eq. (6)). Only the attention module parameters are uploaded to the global server (green arrows), aggregated via sample-weighted FedAVG, and broadcast back to clients (red arrows).

4.7 Inference Pipeline

At inference time, the Llama 3.2 prompt generation pipeline is not used. Instead, a fixed set of K class-level fallback prompts — one per class of the form `"brain MRI showing {class}"` — is encoded once by the frozen BiomedCLIP text encoder to produce a bank of text feature vectors $[\mathbf{T}_1, \dots, \mathbf{T}_K]$. For a test image $\mathbf{x}$, the BiomedCLIP image encoder extracts features $\mathbf{I}$, which are passed through the trained global attention module to produce $\tilde{\mathbf{I}} = a_{\text{global}}(\mathbf{I}) \odot \mathbf{I}$. The cosine similarity between $\tilde{\mathbf{I}}$ and each $\mathbf{T}_c$ is normalised via softmax (Eq. (4)) and the predicted class is:

$$\hat{y} = \arg\max_{c \in \mathcal{Y}} \; P(Y{=}c \mid \mathbf{x}). \tag{14}$$

The entire pass requires only two frozen encoder forward passes and one attention module forward pass. Figure 2 illustrates this pipeline.

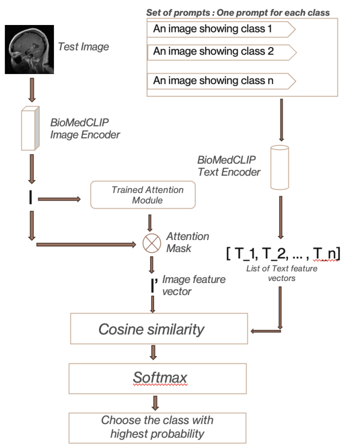

Figure 2. Inference pipeline. Test image → BiomedCLIP encoder → attention module → I'. Class prompts → text encoder → $[\mathbf{T}_1, \dots, \mathbf{T}_K]$. Cosine similarity + softmax → predicted class.

4.8 Evaluation Metrics

We report two complementary metrics for all experiments. **Accuracy (ACC%)** measures the fraction of correctly classified test images and enables direct comparison with FACMIC [19]. **Balanced Accuracy (BACC%)** [2] computes the macro-averaged per-class recall, providing a fairer assessment under class imbalance — the `no_tumor` class is under-represented in the non-iid client distributions, and standard accuracy can be inflated by majority-class bias. Both metrics are computed via `scikit-learn`.

5. Experiments

5.1 Dataset

We evaluate on the Brain Tumor (BT) MRI dataset [1]: four classes (glioma, meningioma, no tumor, pituitary tumor), 2,870 training and 394 test images. Training data is distributed across three clients under non-iid conditions using pre-partitioned FACMIC splits ($\alpha = 0.9$, Dirichlet). `client_3` serves as the unlabelled LMMD target domain and the global test set.

5.2 Implementation Details

All experiments use an **NVIDIA L4 GPU** (24 GB) on Google Colab. BiomedCLIP is loaded via `open_clip` using the checkpoint `microsoft/BiomedCLIP-PubMedBERT_256-vit_base_patch16_224`. The attention module maps 512 → 128 → 512 (BN, LeakyReLU(0.1), Softmax). Key hyperparameters: LR $= 10^{-3}$, batch 32, Adam $(\beta_1, \beta_2) = (0.9, 0.98)$, weight decay 0.02, $\tau = 0.07$, 50 FL rounds, $\lambda = 1$, seed 42.

5.3 Baselines

Our primary baseline is reproduced FACMIC [19] under two hyperparameter settings.

6. Results and Observations

6.1 Reproduced FACMIC Baseline

Reproduced Results

We reproduce FACMIC under: *(i)* **original hyperparamete (HP)** (LR 5×10^{-5}, 50 rounds) as given in the facmic code, and *(ii)* **matched HP** (LR 10^{-3}, 50 rounds) identical to our FABRIC setup, enabling a fair like-for-like comparison.

Table 2. Reproduced FACMIC on `client_3` ($\alpha = 0.9$).

Method	Ep.1	Best	Ep.50	M
FACMIC (orig. HP) – ACC	62.69	**80.71** (ep.27)	79.19	ACC
FACMIC (orig. HP) – BACC	62.73	**79.80** (ep.27)	77.74	BACC
FACMIC (matched HP) – ACC	60.66	**78.68** (ep.43)	78.17	ACC
FACMIC (matched HP) – BACC	60.42	**78.22** (ep.43)	77.44	BACC

FACMIC with matched HP achieves lower best ACC (78.68% vs. 80.71%) and converges later (epoch 43 vs. 27), indicating that CLIP-based FACMIC is sensitive to the learning rate. Both runs serve as baselines for Section 6.3.

6.2 FABRIC: BiomedCLIP + Dynamic Prompts

Figure 5 shows training, validation, and test curves for our full FABRIC framework (BiomedCLIP + dynamic Llama 3.2 prompts, synchronous FL, non-augmented data) over 50 rounds. FABRIC achieves a best test ACC of **84.77**% and BACC of **85.32**% on `client_3` at round 30. Table 3 reports full per-client results.

Table 3. FABRIC results on `client_3` ($\alpha = 0.9$), best checkpoint at round 30.

Split	ACC (%)	BACC (%)
`client_0` (val)	86.30	86.13
`client_1` (val)	78.05	78.33
`client_2` (val)	81.91	80.27
`client_3` (test)	**84.77**	**85.32**

6.3 Comparison

Table 4 compares FABRIC against both FACMIC reproduction settings on the global test set (`client_3`). FABRIC outperforms FACMIC (original HP) by +4.06%

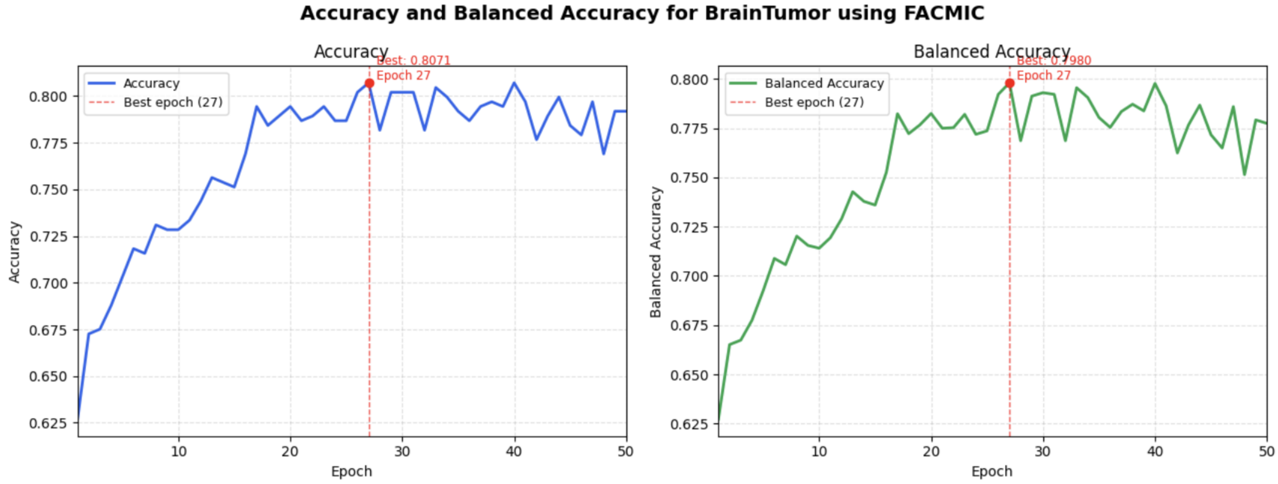

Figure 3. Reproduced FACMIC (original HP): best ACC 80.71%, BACC 79.80% at epoch 27. Steady convergence with minor fluctuations after epoch 20.

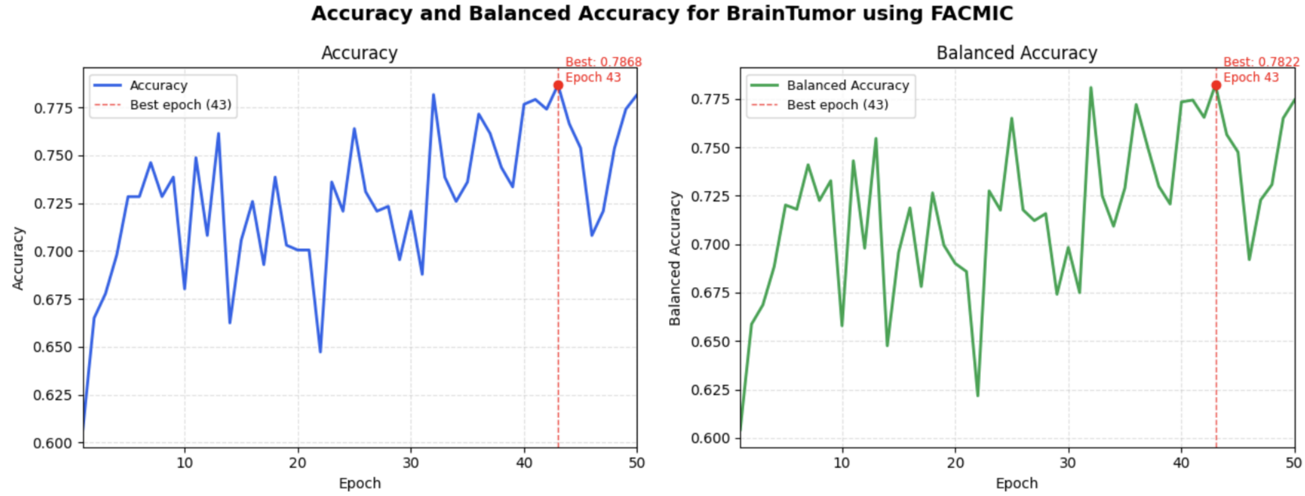

Figure 4. Reproduced FACMIC (matched HP, LR = 10^{-3}, 50 rounds): best ACC 78.68%, BACC 78.22% at epoch 43. Higher variance confirms CLIP-based FACMIC is learning-rate sensitive.

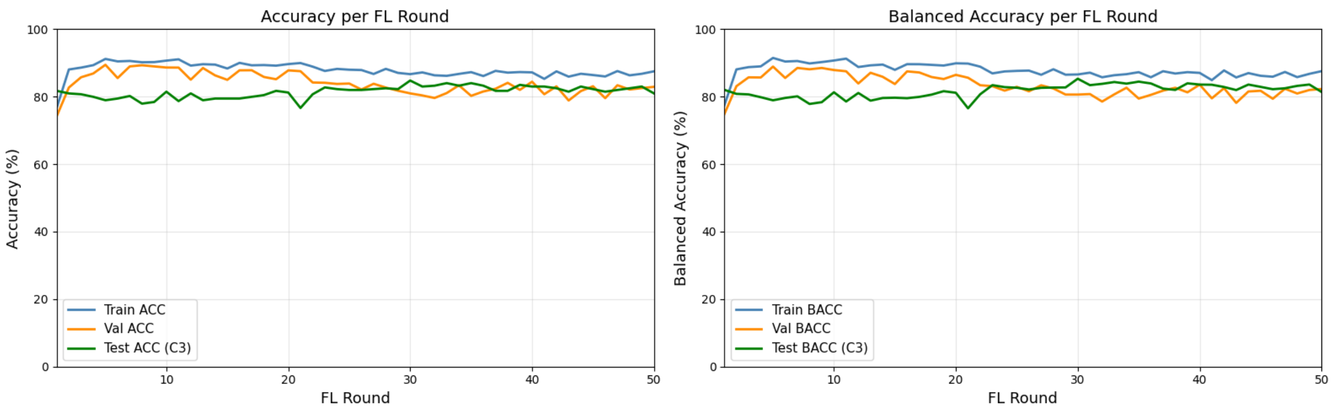

Figure 5. FABRIC (BiomedCLIP + dynamic prompts, sync FL): best ACC 84.77%, BACC 85.32% at round 30. Rapid convergence and stable performance throughout 50 rounds.

ACC and +5.52% BACC, and outperforms FACMIC (matched HP) by +6.09% ACC and +7.10% BACC. The larger improvement over the matched HP setting confirms that the performance gain stems from FABRIC's architectural contributions (BiomedCLIP + dynamic prompts), not from hyperparameter differences. The consistently larger BACC improvement relative to ACC in both cases suggests that FABRIC is particularly effective at improving recall on the minority `no_tumor` class, which standard accuracy tends to underweight.

Table 4. Comparison of FABRIC vs. reproduced FACMIC on `client_3` ($\alpha = 0.9$). Δ_1: vs. original HP. Δ_2: vs. matched HP.

Method	ACC (%)	BACC (%)	Best
FACMIC (original HP) [19]	80.71	79.80	Ep.27
FACMIC (matched HP) [19]	78.68	78.22	Ep.43
FABRIC (Ours)	**84.77**	**85.32**	Rd.30
Δ_1 (vs. original HP)	+4.06	+5.52	—
Δ_2 (vs. matched HP)	+6.09	+7.10	—

7. Ablation Study

7.1 Effect of Asynchronous FL

Our Observations

Over 50 rounds: `client_0` selected 34 times (68%), `client_1` 29 times (58%), `client_2` 37 times (74%), all close to the expected 66.7%. Figure 6 shows training curves; Tables 5–6 report results.

Table 5. Sync vs. Async FL on `client_3` ($\alpha = 0.9$).

Protocol	Clients	ACC	BACC	Rd.
Synchronous	3/3	84.77	85.32	30
Asynchronous	2/3	84.01	84.50	47
Δ	—	−0.76	−0.82	—

Table 6. Per-client validation, async FL, best checkpoint rd.47.

Client	ACC (%)	BACC (%)
`client_0` (val)	82.88	83.15
`client_1` (val)	75.61	76.25
`client_2` (val)	81.91	81.27
`client_3` (test)	**84.01**	**84.50**

Asynchronous FL achieves only ∼0.8% drop in ACC/BACC in exchange for a 33% reduction in per-round communication cost. The best checkpoint arrives later (round 47 vs. 30), consistent with the slower effective update rate when only two clients contribute per round, but convergence remains stable throughout all 50 rounds.

7.2 Effect of Data Augmentation

Our Observations

Figure 7 and Tables 7–8 report results for the augmented synchronous experiment.

Table 7. Effect of augmentation, sync FL, `client_3` ($\alpha = 0.9$).

Setting	ACC	BACC	Rd.
Sync, no aug	84.77	85.32	30
Sync, augmented	84.52	85.04	**10**
Δ	−0.25	−0.28	−20

Table 8. Per-client validation, augmented sync, best at round 10.

Client	ACC (%)	BACC (%)
`client_0` (val)	84.85	84.43
`client_1` (val)	84.09	82.78
`client_2` (val)	89.53	89.24
`client_3` (test)	**84.52**	**85.04**

Augmentation achieves nearly identical test accuracy (−0.25%) while converging 20 rounds earlier. `client_1` gains +8.48% ACC (75.61% → 84.09%), confirming class-aware augmentation alleviates inter-client data imbalance.

7.3 Leave-One-Client-Out Evaluation

Our Observations

Table 9 summarises best-checkpoint results across all four test clients under the LOCO protocol.

Table 9. LOCO evaluation (FABRIC, sync FL, $\alpha = 0.9$).

Test Client	ACC	BACC	Rd.
`client_0`	87.47	87.55	17
`client_1`	88.71	88.61	4
`client_2`	89.78	90.70	9
`client_3`	84.77	85.32	30
Mean	**87.68**	**88.05**	—

Table 10. Per-client validation for each LOCO configuration. C0–C3 denote clients 0–3.

Test	Client	ACC	BACC	Role
C0	C1	82.93	82.50	val
	C2	90.43	91.67	val
	C3	78.95	78.31	val
	C0	**87.47**	**87.55**	*test*
C1	C0	86.99	86.73	val
	C2	86.17	86.97	val
	C3	76.32	75.58	val
	C1	**88.71**	**88.61**	*test*
C2	C0	89.73	89.11	val
	C1	85.37	84.58	val
	C3	68.42	67.86	val
	C2	**89.78**	**90.70**	*test*
C3	C0	86.30	86.13	val
	C1	78.05	78.33	val
	C2	81.91	80.27	val
	C3	**84.77**	**85.32**	*test*

FABRIC achieves mean ACC 87.68% and BACC 88.05% across all four test clients (ACC 84.77%–89.78%). `client_3` yields the lowest result due to its smaller partition size. Consistent generalisation confirms that BiomedCLIP with instance-level dynamic prompts produces robust representations under non-iid federated conditions.

8. Conclusion and Future Work

Conclusion

We presented **FABRIC**, an enhanced federated learning framework that extends FACMIC [19] with a BiomedCLIP backbone, dynamic prompt generation via Llama 3.2 and Ollama, asynchronous FL, and class-aware data augmentation. FABRIC achieves mean ACC 87.68% and BACC 88.05% under LOCO evaluation, outperforming FACMIC by +4.06% ACC and +5.52% BACC (original HP) and +6.09% ACC and +7.10% BACC (matched HP) on the global test set. Asynchronous FL reduces communication cost by one third with only ∼0.8% performance loss. Data augmentation improves per-client balance and accelerates convergence by 20 rounds.

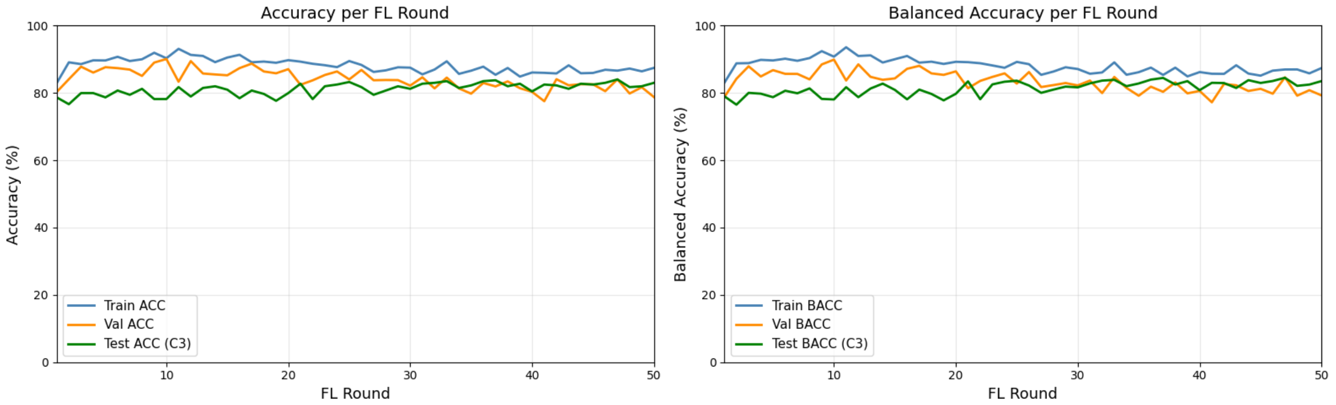

Figure 6. Async FL training curves ($\alpha = 0.9$, 2-of-3 clients/round). Model converges stably and maintains performance close to the synchronous variant.

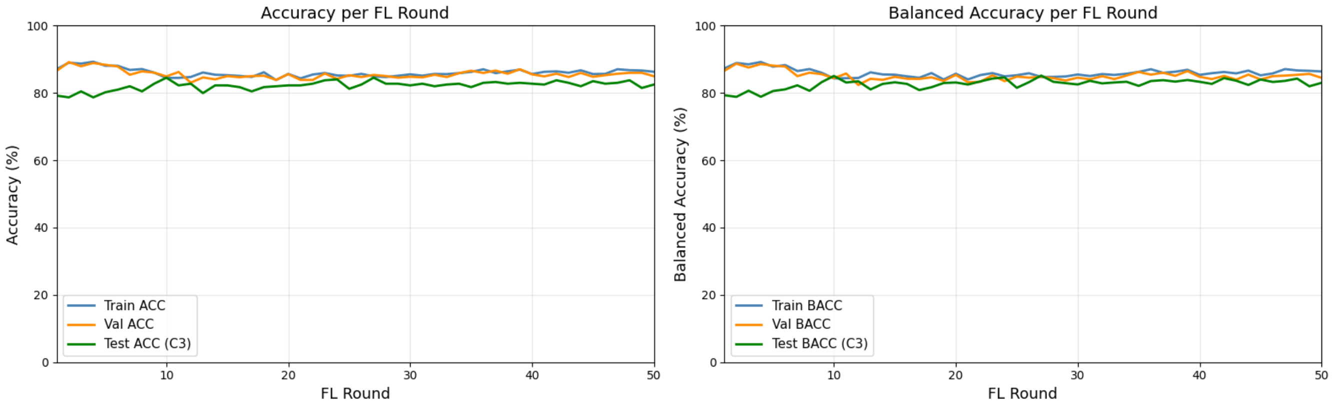

Figure 7. Augmented data experiment (sync FL). Best checkpoint at round 10 vs. round 30 without augmentation. Tight train/val/test alignment indicates reduced overfitting.

Future Work

A natural next step is **Mamba** [6], a state-space model offering linear-time sequence modelling with selective state spaces, attractive for high-resolution medical image encoding in resource-constrained federated settings. Replacing the ViT encoder in BiomedCLIP with VMamba [11] could yield richer spatial features for 3D MRI volumes. Further directions include personalised FL beyond the attention module, extension to histopathology and fundus photography, and incorporating differential privacy into aggregation.

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